

Public Rules, Executed Behavior, and Welfare in a Language-Model Economy

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Abstract

Public natural-language rules are often evaluated as if compliance with the named action were the mechanism that changes welfare. We test that assumption in the Verification Budget Economy, an eight-agent repeated-exchange environment with scarce checks and intrinsically worthless transferable marks. A byte-identical visible-announcement execution intervention shows that blocking feasible Hard-to-Hard check giving lowers welfare within the gift arm by 3.2083 points per account. A prospectively frozen seven-arm study replicates the gift package against no recommendation on Hard-to-Hard swap rate (+0.6070) and welfare (+2.0625), but a zero-call proposal audit finds that the named Easy-to-Hard relation rises only 0.123 above neutral, below a post-hoc 0.25 engagement screen. Any-holder and Hard-partner packages name Hard actors directly, whereas an engaged, mechanically harmful Easy-to-Easy directive also raises Hard giving outside its named relation. The exact money package yields seller proposals but no executed sales, and its named sale contrast does not clear that screen. These results support a layered conclusion: public packages can alter welfare-relevant role-conditional proposal patterns and spill over beyond their named relation, but the present designs do not identify helping-specific scope expansion, a priced- semantic boundary, or a self-enforcing institution. Proposal propagation, physical execution, and welfare are therefore distinct empirical objects.

CCS Concepts

• **Computing methodologies** → **Multi-agent systems**; *Agent / discrete models*; *Natural language processing*.

Keywords

language-model agents, multi-agent systems, public rules, causal mechanisms, welfare, agent-based simulation

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1 Introduction

Public natural-language rules are increasingly used to coordinate language-model teams, markets, and social simulations. Yet an observed change in aggregate outcomes does not reveal which part of the rule-to-outcome chain carried the effect. Agents may propose

the named transaction, respond to an unmentioned role relation, or produce proposals that the environment cannot execute. A study that records only surface compliance can therefore assign welfare to the wrong mechanism.

We study this problem in the Verification Budget Economy (VBE), an eight-account, 24-round random-matching economy in which Hard agents benefit from a partner’s check, Easy agents can salvage an unused check, and transferable marks have no direct or terminal value. Public messages recommend gifting checks or exchanging them for marks, while the engine executes only compatible bilateral transitions. This separation lets us ask three linked questions: does a package engage the relation it names, which execution changes affect welfare, and do explicit actor or partner restrictions constrain the observed response?

We answer these questions in a staged design. W-CO (welfare crowding-out) first discovers a welfare contrast between gift and money talk and exposes an unmentioned H–H channel after the planned gift-versus-sale mechanism gate fails. W-RG (welfare role-generalization channel) then holds the visible public announcement byte-identical and intervenes only after each model response: blocking feasible H–H giving lowers welfare by 3.2083 points per account, with all 12 paired seeds moving in the same direction. W-SGB (welfare semantic-generalization boundary) subsequently varies seven public-message packages under unchanged execution. The gift package replicates on H–H swap rate and welfare, but the zero-call proposal audit shows that its named Easy-to-Hard relation does not clear the post-hoc screen; any-holder and Hard-partner packages name Hard actors directly, while an engaged harmful Easy–Easy directive spills into H–H behavior. The money package produces seller proposals but no executed sales.

The paper makes three contributions. First, it estimates a welfare-relevant execution contrast within the gift arm with a byte-identical visible-announcement intervention rather than inferring mechanism from the public wording. Second, it provides a role-conditional proposal map and a symmetric engagement audit that separates literal support from candidate scope expansion. Third, it establishes an evidence discipline for language-mediated populations: proposal propagation, physical execution, state feedback, and welfare must be measured as distinct layers. The resulting claim is about a causal systems property of a hosted model population, not about private belief, common knowledge, or consciousness.

2 Environment and Rule-to-Outcome Stack

2.1 Verification Budget Economy

VBE contains eight persistent accounts and runs for 24 rounds. In each round, controllers are assigned Hard or Easy roles and randomly matched with meeting probability $q = 0.4$. A successful Hard task pays $R = 3$. A Hard agent’s own check succeeds with probability $p_H = 0.32$; a partner-supplied check succeeds with probability



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Table 1: Expected direct payoff changes for one executed relation relative to no transfer. These are engine accounting identities before downstream matching and payoff draws; a Hard recipient of an E→H transfer retains its own check and therefore also receives its ordinary salvage term.

Relation	Giver	Receiver	Total
H–H swap	+1.83	+1.83	+3.66
E→H gift	−0.50	+2.33	+1.83
H→E gift	−0.96	0.00	−0.96
E–E swap	−0.50	−0.50	−1.00
E→H mark sale	−0.50	+2.33	+1.83

$p_P = 0.93$. At the start of each round, each account receives its round endowment of $B = 1$ check. Every unused inventory check converts to $v = 0.5$ at round end and then expires. Four marks circulate but have no direct or terminal payoff. Replacing an own check with a partner check increases expected Hard-task value by

$$R(p_P - p_H) = 3(0.93 - 0.32) = 1.83. \quad (1)$$

Checks can move through a simultaneous exchange (a *swap*), a one-way E→H gift, or an Easy seller’s mark-contingent transfer matched with a Hard buyer’s offered mark. Controllers propose actions independently; the engine executes only compatible joint transitions. Mean final score per account is the welfare outcome. The principal mechanism outcome is the proportion of H–H meetings that execute a mutual check swap. Because treatment changes holdings and future opportunities, the inferential unit is the complete seed-level population trajectory, not a meeting.

Table 1 makes the transaction accounting explicit. A received check is recorded as a verification source for a Hard agent; it is not added to the recipient’s inventory. In an E→H gift or sale, however, the Hard recipient retains its own check and can still receive the ordinary end-of-round salvage. The prompt exposes the current roles, both inventories, scores, and the last four partner interactions. A sale requires a seller’s mark-contingent proposal and a buyer’s offered mark; a gift requires only the giver’s unconditional proposal. These asymmetric acceptance conditions matter for interpreting zero executed sales.

2.2 Public rules as distributed policy programs

The rule-to-outcome path is

$$\text{message} \rightarrow \text{proposal} \rightarrow \text{execution} \rightarrow \text{state} \rightarrow \text{welfare}. \quad (2)$$

Natural-language evaluations often observe the first two elements and infer the rest. VBE records each layer. We call a message a *distributed policy program* when it changes a population’s mapping from role and local state to proposed action. This is an operational abstraction, not a claim about an internal symbolic representation.

We annotate each directive D as (A_D, B_D, R_D, Q_D) : named actor roles, partner roles, transfer relation, and explicit restrictions such as “only.” This is a descriptive coding scheme, not a formal semantics. A candidate role spillover occurs outside its named actor–partner pairs; a change from a one-way gift to a joint swap is a separate action-level distinction. Confirmatory excess execution rates require both the declared effect-size and error-control gates. Semantic

interpretation additionally needs engagement of the named relation; our later proposal screen checks that prerequisite descriptively.

Exact gift names Easy acting toward Hard, unlike the tested H–H pair. The Hard-partner package allows Hard actors, so H–H fits its role predicate even though a joint swap is not explicitly named. Easy-only both names Easy actors and prohibits Hard giving. Its zero named E→H proposals and compound wording prevent isolation of actor-exclusivity semantics.

3 Experimental Design

3.1 Discovery and execution intervention

W-CO compares historical gift and money packages on 18 fresh paired seeds. The welfare contrast was prospectively evaluated, but the H–H explanation was found only after the planned behavioral-substitution gate failed. W-CO is therefore mechanism discovery, not confirmatory evidence for that channel.

W-RG converts the trace hypothesis into a prospective execution intervention. Twelve fresh seeds each run four paired arms: neutral; standard gift; gift with feasible H–H check giving suppressed after response; and gift with feasible unconditional E→H giving suppressed after response. The three gift arms have byte-identical visible announcements and retain the current meeting’s structured proposal for the local execution transform. They differ only in that narrow post-response transform; subsequent proposals may respond to the changed state. Original intentions are retained because reciprocal Hard giving can turn a candidate one-way E→H gift into a swap before realization. The H–H filter sets `giveCheck=false` if both roles are Hard and the proposer holds a check and proposes to give it. The E→H filter applies the same assignment only to Easy givers facing Hard with `requireChit=false`. Both leave `giveChits` and `requireChit` unchanged. Models are queried separately in each arm.

3.2 Prospective role-boundary test

W-SGB keeps execution unmodified and randomizes seven message packages over 14 fresh paired seeds: neutral; exact gift; gift with the actor restricted to “Easy only”; gift generalized to any check-holder; giving restricted only by a Hard partner; exact money talk; and an Easy–Easy reciprocal-transfer directive. The last is an engaged negative control. When two Easy agents exchange checks, both lose the 0.5 salvage value of an otherwise retained check, so the named execution destroys 1.0 total welfare. A separate E–E swap gate must pass before this arm can support an interpretation about scope.

Co-primary endpoints are seed-level H–H swap rate and mean welfare. Each has eight directional contrasts. Holm adjustment controls familywise error at 0.025 within each endpoint family. Minimum relevant effects are +0.25 for swap rate and +0.5 for welfare. We use MRES (minimum relevant effect size) for these thresholds: the value is a practical-effect gate, so a small significant difference cannot upgrade a scope or welfare claim. The gift, money, and neutral packages are byte-preserved anchors. New packages are not token- or length-matched; W-SGB identifies package-level scope profiles, not a single word or parse. After the frozen run, we borrowed the 0.25 MRES for a descriptive named-proposal engagement screen. This post-hoc zero-call audit qualifies interpretation;

it is not another preregistered hypothesis family. Zero executed sales still cannot establish a priced-semantic boundary.

Table 2 shows the clauses that distinguish the packages. Every directive also states that it is a public statement rather than an engine rule, may be followed or ignored, and that each agent should maximize its own score. The supplement preserves the complete byte-exact messages.

Within each seed, arms share initial holdings, roles, pairings, and pregenerated payoff randomness. Cyclic orders and reversals place every arm twice in each position and balance every pairwise precedence relation 7/7. Strict schemas prevent malformed outputs from becoming idle actions. Each target study is enclosed by an independently specified 14-call pre/post provider-health bracket covering external controls, dominance anchors, prompt hashes, returned model, live catalog, output diversity, entropy, and a metadata fingerprint. W-RG and W-SGB use canonical deepseek-flash in the same provider-labeled V4.1 Flash era. Earlier deepseek-v4-flash results lack comparable returned identities and are not pooled. A bracket bounds an observed service interval; it does not prove immutable weights.

This control was introduced after a serving incident in which 72 distinct prompt hashes returned one identical action pattern. A same-account, same-harness probe on a then-distinct endpoint remained discriminative, and an arithmetic suffix changed the output, showing that the prompt tail was read. We classified that episode as a serving-identity failure rather than a substantive behavioral null. Provider health is consequently evaluated from external controls and metadata independently of each target outcome.

3.3 Estimands and inference

W-RG estimates the paired dynamic total effect of changing an execution contract under a byte-identical visible announcement, with the current meeting’s proposal held fixed only for that local execution transform. W-SGB estimates intention-to-treat effects of complete public-message packages under the unchanged execution contract. Both estimands include downstream feedback through holdings, partner histories, and later responses; role and matching schedules remain paired. They do not decompose a natural indirect effect and do not assume independence among agents or meetings. This population-level treatment assignment follows causal work on interference: when one agent’s treatment changes other agents’ exposures and outcomes, unit-level no-interference estimands are generally unavailable without an exposure model [5, 19].

Formally, let $Y_s(d, e)$ denote the seed- s population outcome under public directive package d and execution policy e , where s fixes the paired initial state, role/matching schedule, and payoff draws. The principal W-RG contrast estimates

$$\frac{1}{12} \sum_s [Y_s(d_{\text{gift}}, e_{\text{standard}}) - Y_s(d_{\text{gift}}, e_{\text{block-HH}})], \quad (3)$$

whereas a W-SGB contrast estimates

$$\frac{1}{14} \sum_s [Y_s(d_1, e_{\text{standard}}) - Y_s(d_0, e_{\text{standard}})]. \quad (4)$$

The first intervenes on execution after the current meeting’s response while allowing subsequent state feedback; it is not an open-loop replay or a natural- mediation estimate. The second intervenes

on a visible package and allows proposals to respond. The two designs therefore answer different causal questions and their estimates are not pooled.

For each contrast we compute an arm difference within every matched seed, then average the paired differences. Reported percentile intervals use 20,000 nonparametric resamples of the paired seed effects with a fixed analysis seed. Directional p -values enumerate every sign assignment of the absolute paired effects under a zero-centered sign-symmetry null. W-SGB applies Holm adjustment separately to its eight H–H and eight welfare hypotheses. A gate passes only when every frozen pair is present, the mean reaches its MRES, and the adjusted p -value is at most 0.025. Thus statistical significance below the declared magnitude—as for gift minus negative—cannot upgrade the verdict.

W-RG completes 3,680 target calls and W-SGB 7,602, with zero retained API, schema, or parse failures; their two pre/post brackets add 56 calls reported separately. Counts of meetings and transfers describe mechanisms, whereas inference remains at the seed-level population unit.

4 Results

4.1 Most candidate one-way gifts resolve as swaps

In W-CO, gift talk exceeds money talk by 2.3333 mean score points (95% bootstrap CI [1.7292, 2.9306], one-sided exact $p = 0.00001526$), but its planned mechanism gate fails: the $E \rightarrow H$ gift-rate difference is only 0.03158, below the 0.25 threshold, and neither arm produces a mark sale. Post-hoc traces instead show H–H swaps rising from 27/154 under money talk to 138/154 under gift talk. The discovery therefore identifies a welfare-relevant channel, not a clean one-way-gift substitution.

W-RG then isolates that channel within the gift package (Table 3). Standard gift produces 83 swaps in 107 H–H meetings; H–H blocking reduces the count to zero. Suppressing H–H giving lowers welfare by 3.2083 points per account, with all 12 paired seeds positive. Retaining H–H circulation while suppressing one-way $E \rightarrow H$ realization exceeds the converse execution regime by 2.7604, again 12/12 positive. Figure 2 contrasts the named $E \rightarrow H$ intention path with the unmentioned H–H circulation and diagrams the post-response execution fork.

Low realized one-way gifting does not imply weak Easy intention. Standard gift contains 26 feasible original $E \rightarrow H$ gift intentions; 25 meet unconditional Hard check-giving and resolve as swaps, leaving one realized one-way gift. Most candidate one-way intentions are therefore absorbed into reciprocal swaps in the realized trace; this describes the outcome pattern rather than a mediation mechanism. The H–H intervention removes 102 Hard solves. At three points per solve this accounts for 306 aggregate points against 308 observed aggregate welfare points. This is dependency accounting, not an independent mediation estimate: blocking changes later state, holdings, histories, and responses. The identified quantity is the dynamic total effect of disabling the H–H channel. Because W-RG has no neutral-with-H–H-blocked arm, it does not identify the fraction of the gift-minus-neutral welfare effect carried by H–H circulation. That directive-by-execution interaction remains a planned follow-up.

Table 2: W-SGB message packages as descriptive named-support annotations. Q_D records an explicit closure condition; $\emptyset$ denotes no such condition.

Arm	Distinguishing public clause (abridged)	Annotation (A_D, B_D, R_D, Q_D)
Neutral	“announces no exchange recommendation”	$\emptyset$
Gift exact	“an Easy agent gives ... to a Hard agent”	$(\{E\}, \{H\}, \{gift\}, \emptyset)$
Easy only	“only an Easy agent gives ...; A Hard agent does not”	$(\{E\}, \{H\}, \{gift\}, \{actor-exclusive\})$
Any holder	“any agent who holds a leftover check gives it”	$(\{E, H\}, \{H\}, \{gift\}, \emptyset)$
Hard partner	“an agent gives ... only when the current partner is Hard”	$(\{E, H\}, \{H\}, \{gift\}, \{partner-exclusive\})$
Money exact	“a mark is accepted as payment for one leftover check”	$(\{E\}, \{H\}, \{sale\}, \emptyset)$
E–E negative	“when two Easy agents meet ... each gives ... to the other”	$(\{E\}, \{E\}, \{swap\}, \emptyset)$

Public messages, execution rules, and welfare

Discovery, execution intervention, and prospective message comparison in the same economy.

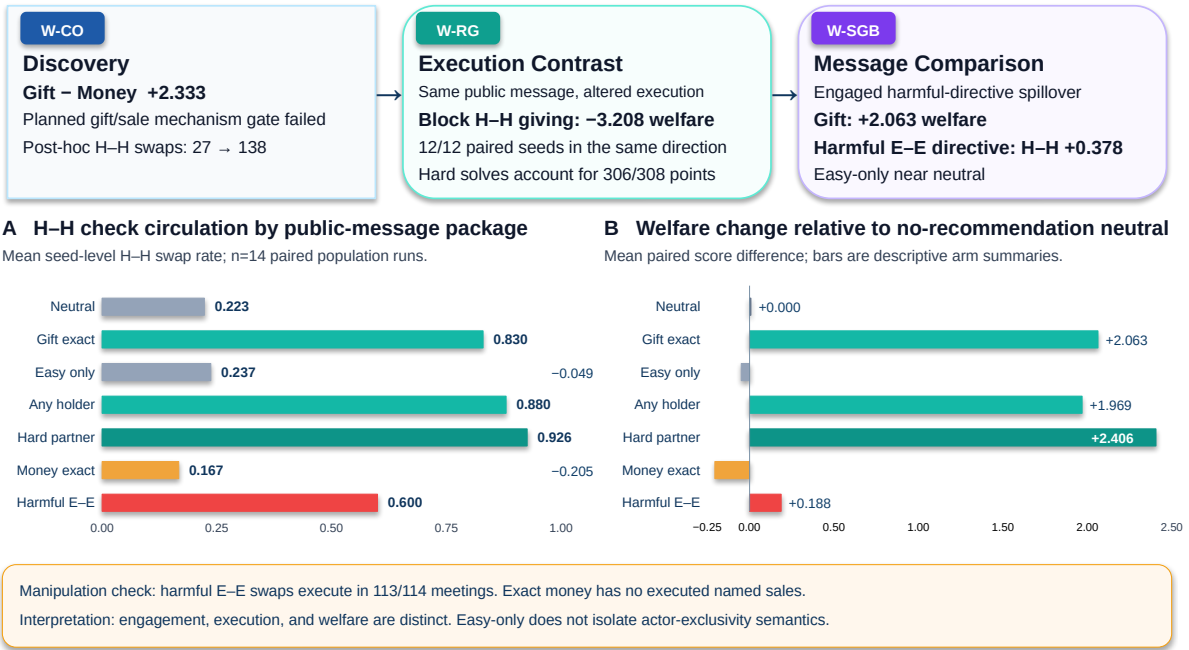


Figure 1: Discovery, byte-identical visible-announcement execution contrast, and the prospective role-boundary study. The named Easy-to-Hard gift rarely realizes as a one-way transfer; Hard reciprocity converts most candidate gifts into swaps, and welfare falls when feasible Hard–Hard circulation is blocked under the gift package.

Table 3: W-RG prospective paired contrasts.

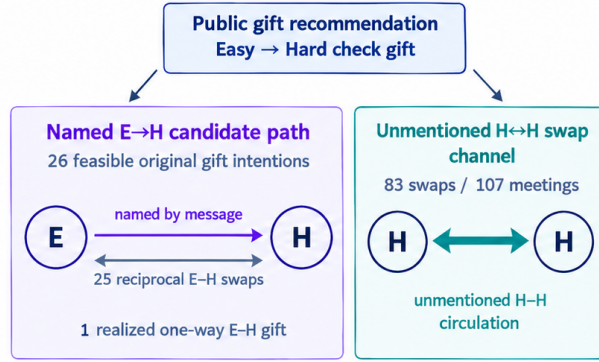
Contrast	Mean Δ	95% bootstrap CI	Exact p	Frozen gate
Gift – neutral welfare	1.6823	[.8542, 2.4635]	.00342	pass
Gift – H–H-blocked welfare	3.2083	[2.4688, 3.8906]	.00024	pass; 12/12 positive
E→H-blocked – H–H-blocked welfare	2.7604	[2.0938, 3.4115]	.00024	pass; 12/12 positive
Gift – H–H-blocked swap rate	0.7525	[.6327, .8615]	.00024	pass

4.2 Gift replicates; helping specificity remains unresolved

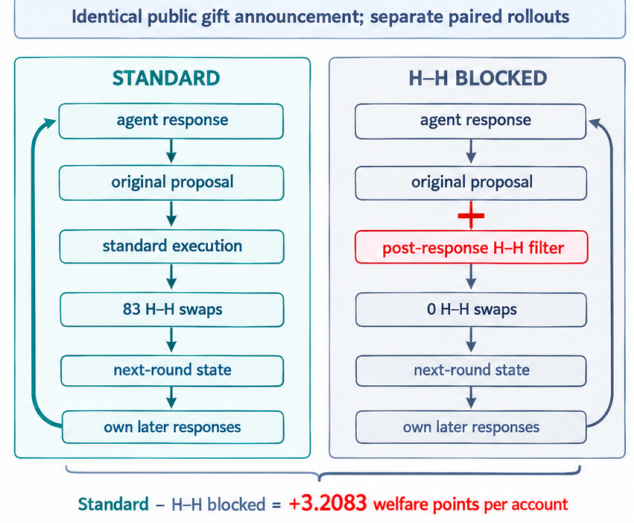
W-SGB completes all 98 cells. Exact gift increases H–H swap rate over neutral by 0.6070 and welfare by 2.0625; both are positive in

14/14 seeds and have Holm-adjusted $p = 0.0004883$ (Table 5). The gift reference thus replicates on both endpoints. Because neutral announces no recommendation, this comparison estimates the total effect of the gift directive’s presence and content together; it is not

A STANDARD-GIFT TRACE



B W-RG EXECUTION CONTRAST



102 fewer Hard solves; 306/308 aggregate score-point gap accounted

Figure 2: The named candidate path and the unmentioned channel. With the revised paired-rollout schematic, identical public announcements feed the standard and H-H-blocked execution paths; the solve accounting is dependency accounting rather than an independent mediation estimate.

by itself gift-specific. The zero-call audit in Table 4 then checks whether each package actually raises its named proposal relation. The same exact-gift arm also records 99 H→E gifts (versus 3 in neutral); because each has negative direct accounting, the package does not have a uniformly welfare-positive realized relation. Neutral itself contains 27/125 H-H swaps; the uptake timing is descriptive and does not establish norm formation or monotone adoption.

The audit changes the scope interpretation. Exact gift raises H-H giving strongly, but its named E→H proposal rate is only 0.1229, with a seed-level mean difference of 0.1229 from neutral, below the 0.25 screen. The Easy-only package never proposes its named E→H gift and therefore cannot identify actor exclusivity. Any-holder and Hard-partner packages clear the proposal screen, but their actor support explicitly includes Hard, so their H-H responses are inside the named support. The harmful E-E package is both engaged and outside its named E-E support when it raises H-H giving. The result is evidence that public packages alter role-conditional proposal patterns and can produce engaged cross-role spillover; structured scope expansion, role misbinding, and generic transfer priming remain observationally equivalent.

As an environment-only calibration, we replayed the 14 seeds with scripted policies. Never transferring checks scored 54.241 on average, first-best Hard-directed transfer 63.188, and always giving 60.813. A model-free surrogate sampled from the exact-gift proposal map scored 56.683; suppressing its H-to-E component raised this to 57.853. These yardsticks are descriptive, not new model evidence: they expose welfare-relevant waste channels but do not identify why the model selected them.

4.3 Near-total compliance with a harmful directive

The negative directive adds no engine rule, reward, monitoring, or sanction, and explicitly says that agents may ignore it. Yet it executes 113 E-E swaps in 114 opportunities. Each named execution removes 0.5 points from each Easy agent’s private salvage and destroys 1.0 total welfare. The zero-call proposal audit reports a seed-level mean E-E gift-proposal rate of 0.991; the corresponding pooled directed count is 227/228. Separately, the frozen execution engagement gate reports a seed-level mean E-E swap-rate contrast of 0.9821 (113/114 pooled meeting swaps), with 95% CI [0.9464, 1.0000] and adjusted $p = 0.0000610$. These are distinct proposal and execution metrics. Despite naming only E-E transactions, the directive raises H-H swap rate over neutral by 0.3777. The payoff table shows a direct cost of 1.0 per E-E swap, but this design cannot tell whether agents knowingly accept that cost or fail to represent the salvage consequence. The latter question is reserved for a payoff-salience replay probe.

Near-total named compliance therefore does not establish welfare alignment. The same package also induces potentially useful H-H circulation, and its net welfare difference from neutral is only 0.1875, with a confidence interval crossing zero and below the 0.5 threshold. A public rule can be behaviorally contagious while imposing a per-transaction loss; aggregate welfare damage is not established here.

No W-SGB arm yields an executed mark sale, although E→H seller proposals are present. Descriptively, its H-H effect is -0.0560 (bootstrap CI $[-0.1259, -0.0009]$) and its welfare effect is -0.2054

Table 4: Post-hoc zero-call proposal audit. The named column is the package’s seed-level named-relation proposal rate minus neutral; the three role columns are seed-level means for the indicated directed cell. The displayed 0.25 MRES is a descriptive screen applied after the frozen run, not a preregistered inferential gate.

Arm	Named Δ	E→H	H→H	H→E	MRES screen
Neutral	—	0.000	0.261	0.012	reference
Gift exact	0.123 (E→H)	0.123	0.906	0.427	fail
Easy only	0.000 (E→H)	0.000	0.353	0.054	fail
Any holder	0.365 (actor→H)	0.113	0.927	0.420	pass [†]
Hard partner	0.327 (actor→H)	0.020	0.963	0.066	pass [†]
Money exact	0.063 (E→H sale)	0.000	0.191	0.000	fail
Harmful E→E	0.991 (E→E)	0.013	0.737	0.130	pass

[†]Hard actors are inside the named support, so these are literal-support responses.

Table 5: Selected W-SGB contrasts. CIs are seed-level bootstrap intervals; p_H is Holm-adjusted within each eight-contrast endpoint family.

Contrast	H-H swap rate			Mean welfare		
	Δ	95% CI	p_H	Δ	95% CI	p_H
Gift – neutral	.6070	[.5097,.7140]	.00049	2.0625	[1.4286,2.8125]	.00049
Gift – money	.6631	[.5681,.7551]	.00049	2.2679	[1.6696,2.9598]	.00049
Gift – negative	.2293	[.1049,.3506]	.00537 [†]	1.8750	[1.3482,2.4330]	.00049
Money – neutral	-.0560	[-.1259,-.0009]	.96875	-.2054	[-.3393,-.0714]	.99609
Negative – neutral	.3777	[.2875,.4745]	.00049	.1875	[-.3393,.7098]	.50916
Gift – Easy only	.5923	[.4820,.6974]	.00049	2.1116	[1.4643,2.8125]	.00049
Any holder – neutral	.6576	[.5488,.7713]	.00049	1.9688	[1.2946,2.6964]	.00049
Hard partner – neutral	.7034	[.5956,.8100]	.00049	2.4063	[1.7589,3.1071]	.00049

[†]Directional significance does not establish the prespecified 0.25 minimum relevant effect; the interval crosses that practical-effect boundary.

([−0.3393, −0.0714]) relative to neutral. The registered tests were one-sided for positive effects, so these negative directions were not prospectively tested and are not upgraded. They motivate an engagement-gated replication of possible suppression, but cannot distinguish “priced-exchange semantics do not expand” from a failure to match seller and buyer proposals. The current data therefore support no priced-semantic boundary claim.

4.4 Boundary arms do not isolate actor exclusivity

The “Easy only” arm has mean H-H swap rate 0.237, near neutral’s 0.223, and mean welfare 0.049 below neutral. Exact gift exceeds it by 0.5923 in swap rate and 2.1116 in welfare, clearing both frozen gates. Descriptively, the Easy-only variant remains near neutral, but its explicit prohibition and zero named E→H proposal rate prevent a quantifier-only interpretation. By contrast, “any check-holder” and “only when the partner is Hard” retain strong H-H circulation and welfare effects; the latter explicitly includes Hard actors in its named support. Figure 3 summarizes the tested role relations and their H-H responses, including the harmful directive and the unmatched money package.

5 Discussion

5.1 What the evidence identifies

Taken together, the studies identify a layered causal object rather than a single semantic effect. W-RG shows that, within the gift arm,

changing the post-response execution contract to block feasible H-H giving lowers welfare by 3.2083 points per account. Because no neutral-with-H-H-blocked arm was run, this dynamic total effect does not identify the share of the gift-versus-neutral contrast carried by H-H circulation or establish necessity. W-SGB shows that the same package changes H-H behavior relative to neutral, but the zero-call audit prevents us from treating that response as a demonstrated helping-specific expansion, because the named E→H relation does not clear the post-hoc screen. The any-holder and Hard-partner packages name Hard actors directly, and the harmful E→E package demonstrates engaged spillover without a helping interpretation. These alternatives remain observationally distinct in meaning but not in the present data.

The best-supported abstraction is therefore a *distributed policy program*: public text changes a population’s mapping from local roles and state to proposals, and the engine determines which proposals become state transitions. This is more than an ignored string, but less than a social institution with common belief, legitimacy, consent, stable financing, or open-ended equilibrium support. Reserving “institution” for that stronger object keeps the causal finding useful without assigning the model population beliefs or legitimacy that were not measured.

5.2 Implications for multi-agent system design

The design implication is an audit requirement. Natural-language rules should be versioned like code: systems should log exact text,

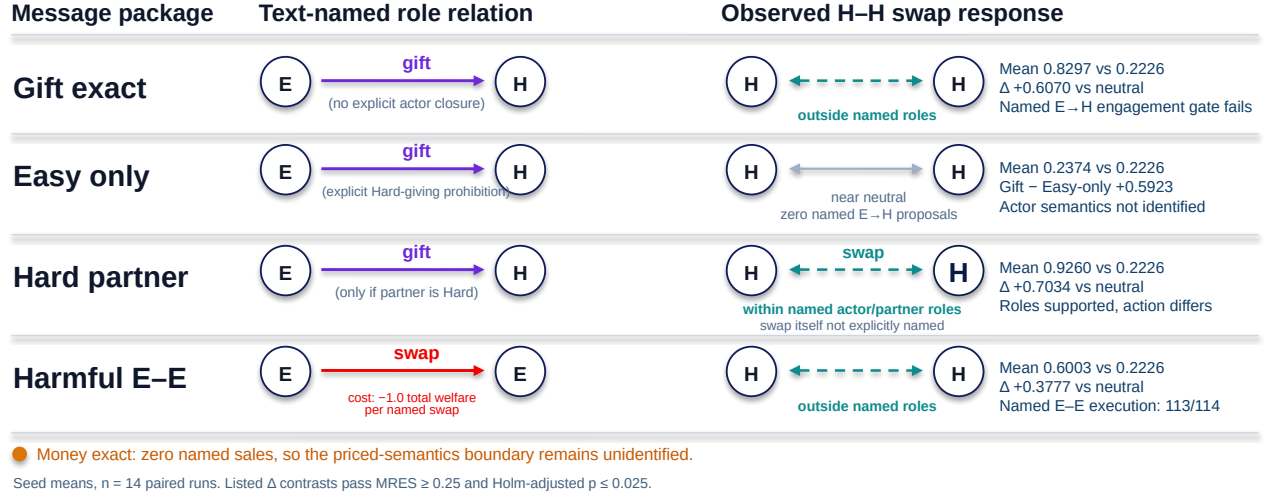


Figure 3: Tested role relations and package-level response. The revised schematic reports each tested Hard–Hard rate beside its neutral contrast and makes the frozen thresholds explicit; the exact-money package did not engage, so it does not identify a priced-semantics boundary.

actor and beneficiary scope, model-service identity, proposal output, execution contract, and welfare transition. Evaluations should first test whether the named relation engages, then probe unmentioned interactions with an engaged undesirable directive; otherwise a helping-specific interpretation can survive because its comparator did nothing. Proposal and execution should remain separate audit layers, and the final report should distinguish named compliance, cross-role spillover, realized transitions, state feedback, and welfare. Optimizing only the first two can select a rule that is legible and behaviorally contagious while still inducing a per-transaction loss; aggregate welfare damage must be established separately.

6 Limitations

The primary results concern one hosted endpoint, one provider-labeled era, and one finite-horizon economy. Temperature zero and healthy pre/post brackets do not establish immutable weights or cross-model generality. W-SGB identifies whole-message packages rather than tokens or parses. The unengaged money arm cannot locate a priced-semantic boundary; one frozen repair attempt ended incomplete after a provider transport failure and yields no inference. The only cross-family concordance is three unpaired, unbracketed historical grok-4.5 seeds and is too weak to count as replication. More usefully, that earlier interface produced 24 mark sales in 44 opportunities while the current one produces none, demonstrating context dependence without identifying an era effect.

Treatment changes population trajectories, so seed-level runs are the units and the work does not identify natural-mediation shares. The economy fixes price, mark supply, horizon, and task technology, omitting issuance, price discovery, counterfeiting, currency competition, and participant-funded enforcement. Frozen account-level summaries are exploratory diagnostics only and do

not identify fairness or legitimacy. The studies observe public messages, proposals, executions, and outcomes, not private belief, common knowledge, consent, or consciousness. Protocols were hash-frozen locally before target calls but not externally registered. The H–H endpoint was selected after W–CO’s mechanism failure; fresh-seed prospective tests assess this selected phenotype, not the original mechanism hypothesis or an unselected program-wide claim. Future work should use fresh seeds and matched directive-force, reciprocity, actor-scope, beneficiary-scope, and engagement-gated price variants, then replicate on a verifiably distinct model family.

7 Related Work

Institutional statements and operative rules. Institutional grammar decomposes rules into attributes, deontic operators, aims, conditions, and sanctions [12]; formal accounts distinguish institutional power from permission and physical possibility [20], and constitutive rules from regulative norms [9]. Normative multi-agent-systems work then represents role-indexed obligations, permissions, monitoring, enforcement, and institutional change explicitly [14, 17, 21]. These traditions specify what a rule means or authorizes. VBE asks a complementary execution question: when the rule is natural language interpreted by model policies, do its realized actor, beneficiary, action, and condition scopes match the written ones? Separating model proposal from engine execution operationalizes the distinction between a normative statement and a state transition.

Institutions, conventions, and norm emergence. Accounts of convention and common knowledge separate shared observation from higher-order epistemic states [24, 35], while experiments study spontaneous convention formation and tipping [10, 11]. Social-law synthesis studies top-down constraints [40]; work on norm emergence studies bottom-up convergence through social learning, interaction, propagation, and norm life cycles [28, 36, 37, 39]. VBE’s

public artifact is top-down and its response is immediate, so we do not label it bottom-up emergence, internalization, or common belief. Searle’s collective status functions and Ostrom’s participant-fitted governance similarly set a higher bar for “institution” than behavioral compliance alone [30, 38].

Generative social simulation. Agent-based computational social science explains macro-patterns by specifying interacting actors and generating population dynamics [13, 26]. LLM work extends this program to believable agents, simulated human samples, replications of human-subject studies, and interactive social-intelligence evaluation [2–4, 31, 43]. Concordia provides a componentized generative agent-based modeling framework with a grounded action resolver [41]; economic simulations such as Homo Silicus and EconAgent use language models to instantiate heterogeneous economic actors and market dynamics [18, 25]. These systems establish the usefulness of language-mediated simulation, but they do not by themselves identify whether a public rule engages its named relation before a macro-level outcome is interpreted. Our target is therefore not human fidelity or believability, but a causal systems property of a hosted model population: how a versioned public artifact changes proposals, executable transitions, cross-role spillovers, and welfare under a fixed engine.

Language-model collectives. Recent studies examine coordination, cooperation, convention formation, legal structures, and economic interaction in LLM populations [1, 15, 27, 33, 34, 42, 44]. GovSim studies sustainable cooperation in a common-resource society, and Welfare Diplomacy benchmarks general-sum cooperation under strategic pressure [29, 32]. These benchmarks make collective welfare and failure modes visible, but their aggregate outcomes do not separate named proposal engagement from the execution channel that carries the outcome. The naming-game debate further shows that collective behavior can be observationally compatible with training-data reproduction [8, 16]. We therefore do not infer emergence from novelty. Our contribution is an identification design: a byte-identical execution intervention locates a welfare-relevant contrast within the gift arm, and an engaged harmful control tests cross-role spillover without equating surface compliance with mechanism or value.

Causal effects in interacting populations. General-interference frameworks make outcomes functions of other units’ treatments or exposures and require estimands matched to that dependence [5, 19]. In VBE, early actions change holdings, later partners’ opportunities, and future prompts, so the inferential unit is the paired seed-level population trajectory rather than a meeting or account. This supports dynamic total effects, not individual direct effects or natural mediation shares. Correlated-equilibrium theory formalizes how shared information can coordinate pure actions without explicit randomization [6, 7]; our design does not infer a common prior or equilibrium. Search-theoretic monetary models motivate the matching and token environment [22, 23], but fixed supply, horizon, and prices preclude a monetary-equilibrium claim. These causal and economic frameworks motivate our separation of proposal, execution, and welfare, but the contribution here is a population-level identification design rather than a training or calibration method.

8 Conclusion

A public sentence can change a language-model population without directly specifying the welfare-carrying transition. In VBE, a byte-identical visible-announcement execution intervention shows that blocking feasible H–H giving lowers welfare by 3.2083 points per account within the gift arm; without a neutral-with-blocked comparison, this does not establish a necessary channel or a mediation share for the gift-versus-neutral effect. The proposal audit shows that the named E→H relation fails the post-hoc screen, so no helping-specific expansion claim is identified. The harmful E–E directive spills into H–H behavior, while Easy-only remains near neutral and cannot identify actor exclusivity; money is unengaged and sets no priced-semantic boundary. The contribution is an audit rule: separate proposal, execution, feedback, and welfare.

AI assistance and artifacts. OpenAI Codex (GPT-5-based agent; exact backend revision unavailable) assisted hypothesis/design iteration, code, analysis, and editing. Authors remain responsible; no AI system is an author. Frozen protocols, traces, and full hashes are retained locally. The AI disclosure currently contains representative prompts; complete contextualized prompt records, author verification, and the anonymized supplement must be completed before submission.

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